DEEPAK KATUKURI

AI / ML ENGINEER · LLM & RAG · EVALUATION



CONTACT

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TECHNICAL SKILLS

Languages: Python, SQL, PL/SQL, Core Java, C++

LLM & RAG: Retrieval-Augmented Generation, LLMs, LangChain, LangGraph, hybrid retrieval (BM25 + dense, RRF), cross-encoder reranking, vector databases (pgvector, Chroma), CLIP, DistilBERT, sentence-transformers, contrastive learning, embeddings, Ollama

Deep Learning: PyTorch, TensorFlow, CNNs, fine-tuning, transformers, HuggingFace, timm, EfficientNet

ML & Data: Scikit-Learn, model evaluation, benchmarking, classification, ensembles, NumPy, Pandas, Matplotlib, Seaborn

Cloud & DevOps: AWS (EC2, RDS, VPC, Elastic IP), Kubernetes (K3s, HPA, Ingress, Secrets), Docker, Docker Compose, GitHub Actions, CI/CD, Vercel, Render

Full-Stack: FastAPI, Flask, React, Vite, Tailwind CSS, PostgreSQL, MySQL, SQLAlchemy, Alembic, JWT, Recharts

Tools: Git/GitHub, pytest, Jupyter, Linux (Ubuntu)

EDUCATION

MSc Data Science
University of Europe, Germany
Sep 2025 – Present

BTech in Computer Science & Engineering
MLRIT, India
Jun 2018 – Jul 2022

CERTIFICATIONS

PL/SQL (Database Management)
Core Java (Software Development)
OneShield (P&C Insurance)

LANGUAGES

English – Fluent
German – A1/A2 (learning)

PROFILE

AI/ML Engineer and Data Science MSc candidate, with hands-on LLM, RAG and embedding work and 2.5 years of production software engineering. Built a hand-rolled hybrid RAG retriever and benchmarked it against a LangChain rebuild on a 55-question golden set, then added a corrective loop that refuses questions the corpus cannot answer. Trained a PyTorch CLIP dual-encoder from scratch for image-to-recipe retrieval and evaluated it by median rank. Looking for an English-speaking working student (Werkstudent) Machine Learning role in Germany.

PROJECTS

RAG Pipeline: Hand-Rolled Retrieval to Corrective RAG (CRAG)

Python · LangChain · pgvector · BGE-M3 · Corrective RAG · github.com/Deepak589/RAG_pipeline

- Built retrieval from scratch through 6 gated stages — hybrid BM25+dense, then benchmarked against a LangChain rebuild on a 55-question golden set (caught a silent 0.06-recall regression from a default tokenizer) — and added a corrective RAG loop that re-retrieves on a bad match and refuses when the corpus can't answer, reaching 100% correct refusal and 69.5% end-to-end correctness.

Food Image-to-Recipe Retrieval: Multimodal Deep Learning

PyTorch · CLIP ViT-B/32 · DistilBERT · InfoNCE · MSc final project

- Trained a PyTorch CLIP ViT-B/32 + DistilBERT dual-encoder with symmetric InfoNCE into a shared 1024-d space, reaching a validation median rank of 10/200 on a 13.5K-image Kaggle dataset while cutting training time ~10x by caching CLIP image features.

JobPilot: Multi-Agent Job Search & CV Tailoring Pipeline

Python · LangGraph · Chroma · Typst · github.com/Deepak589/Job_hunt_Agentic_AI

- Built a 16-node LangGraph pipeline that diagnoses a job description, rewrites CV bullets, and blocks any claim without evidence in a Chroma-backed profile store — checkpointed for crash recovery, cost-tracked per run, and held honest by 238 pytest tests across 36 files.

CloudNotes: Kubernetes & AWS Deployment Platform

AWS EC2 + RDS · K3s · GitHub Actions · Traefik · MSc cloud computing, team of 3 · github.com/Deepak589/cloudnotes

- Replaced a team's manual deploys with a GitHub Actions pipeline to AWS Kubernetes (K3s) — build, push, SSH, zero-downtime rollout — autoscaling 2 to 5 pods at 60% CPU; owned the Kubernetes, CI/CD and AWS layers on a 3-person team.

RA Nutrition & Lifestyle App: Shipped Full-Stack Product

React · FastAPI · PostgreSQL · Docker · live at ra-project-indol.vercel.app · github.com/Deepak589/RA_project

- Shipped a full-stack health app — React/Vite frontend on Vercel, FastAPI + PostgreSQL backend on Render — that turns daily meal and symptom logs into a rule-based recommendation and a weekly analytics dashboard, backed by a 99-test pytest suite.

EXPERIENCE

Software Engineer

Mar 2022 – Sep 2024

ValueMomentum · clients: Hiscox, Philadelphia Insurance

- Migrated full policy, claims, premium and customer history for Philadelphia Insurance's Surety Bonds line from a legacy policy admin system into OneShield PAS. Wrote and validated the PL/SQL migration scripts across customer, policy (PID), claims and premium tables.
- Root-caused a recurring data-integrity defect where claims data for pre-2019 Surety bonds under one bond type loaded empty. The OneShield object copy from the stored object to the loading object failed silently when another bond shared that bond type under a different bond ID. Kept records current with a weekly backfill script while a permanent fix was pending.
- Resolved 100+ production defects across two carrier platforms, Hiscox (P&C insurance) and Philadelphia Insurance (surety bonds). Turned underwriting and claims requirements into technical specs, and ran QA/UAT test cases and CI/CD deployments across the full policy lifecycle.

ML Intern

Jun – Jul 2019

SmartBridge × IBM

- Built a lung-cancer risk-prediction ensemble (Random Forest, KNN, Gradient Boosting, Neural Network) and deployed it on IBM Watson ML. Presented the results to non-technical reviewers with Matplotlib and Seaborn charts.